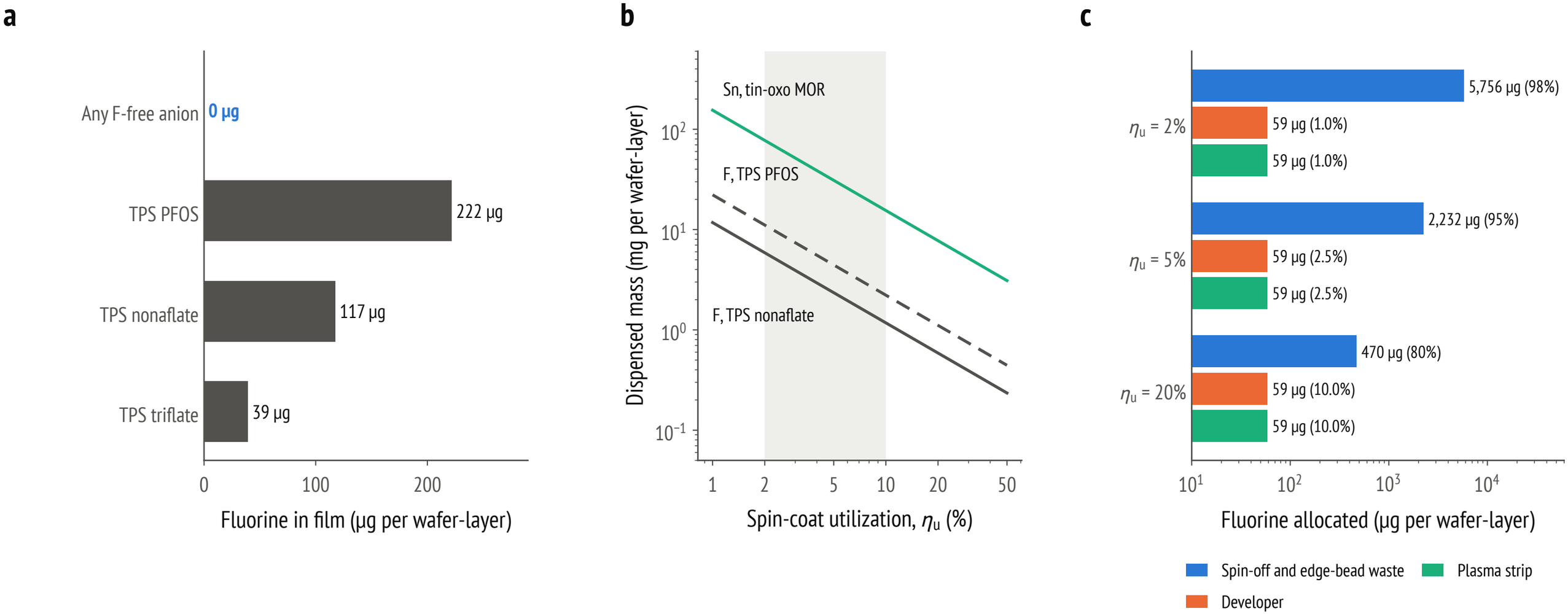
a
Any F-free anion
0 µg
TPS PFOS
222 µg
TPS nonaflate
117 µg
TPS triflate
39 µg
0
100
200
Fluorine in film (µg per wafer-layer)
b
Sn, tin-oxo MOR
F, TPS PFOS
F, TPS nonaflate
Dispensed mass (mg per wafer-layer)
10^2
10^1
10^0
10^{-1}
1
2
5
10
20
50
Spin-coat utilization, η_u (%)
c
η_u = 2%
5,756 µg (98%)
59 µg (1.0%)
59 µg (1.0%)
η_u = 5%
2,232 µg (95%)
59 µg (2.5%)
59 µg (2.5%)
η_u = 20%
470 µg (80%)
59 µg (10.0%)
59 µg (10.0%)
10^1
10^2
10^3
10^4
Fluorine allocated (µg per wafer-layer)
Spin-off and edge-bead waste
Plasma strip
Developer